

Θέματα Εξαμηνιαίας Εργασίας Φωνής και Γλώσσας 2025-26

Διδάσκων: Α. Ποταμιάνος

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AP — Speech Applications

[AP-1] EmoVoice: LLM-based Emotional Text-To-Speech Model with Freestyle Text Prompting

Link: <https://arxiv.org/abs/2504.12867>

Human speech goes beyond the mere transfer of information; it is a profound exchange of emotions and a connection between individuals. While Text-to-Speech (TTS) models have made huge progress, they still face challenges in controlling the emotional expression in the generated speech. In this work, we propose EmoVoice, a novel emotion-controllable TTS model that exploits large language models (LLMs) to enable fine-grained freestyle natural language emotion control, and a phoneme boost variant design that makes the model output phoneme tokens and audio tokens in parallel to enhance content consistency, inspired by chain-of-thought (CoT) and chain-of-modality (CoM) techniques. Besides, we introduce EmoVoice-DB, a high-quality 40-hour English emotion dataset featuring expressive speech and fine-grained emotion labels with natural language descriptions. EmoVoice achieves state-of-the-art performance on the English EmoVoice-DB test set using only synthetic training data, and on the Chinese Secap test set using our in-house data.

[AP-2] SAILER: Developing a Top-tier Framework in Naturalistic Conditions Challenge for Categorized Emotion Prediction

Link: <https://arxiv.org/abs/2505.22133>

Speech emotion recognition (SER), particularly for naturally expressed emotions, remains a challenging computational task. Key challenges include the inherent subjectivity in emotion annotation and the imbalanced distribution of emotion labels in datasets. This paper introduces the SAILER system developed for participation in the INTERSPEECH 2025 Emotion Recognition Challenge (Task 1). The challenge dataset, which contains natural emotional speech from podcasts, serves as a valuable resource for studying imbalanced and subjective emotion annotations. Our system is designed to

be simple, reproducible, and effective, highlighting critical choices in modeling, learning objectives, data augmentation, and engineering choices. Results show that even a single system (without ensembling) can outperform more than 95% of the submissions, with a Macro-F1 score exceeding 0.4. Moreover, an ensemble of three systems further improves performance, achieving a competitively ranked score.

LLM — Architecture & Optimization

[LLM-1] HC + mHC: Hyper-Connections and Manifold-Constrained Hyper-Connections

Link 1: <https://arxiv.org/abs/2409.19606>

Link 2: <https://arxiv.org/abs/2512.24880>

HC: We present hyper-connections, a simple yet effective method that can serve as an alternative to residual connections. This approach specifically addresses common drawbacks observed in residual connection variants, such as the seesaw effect between gradient vanishing and representation collapse. Theoretically, hyper-connections allow the network to adjust the strength of connections between features at different depths and dynamically rearrange layers. mHC: Recently, studies exemplified by Hyper-Connections (HC) have extended the ubiquitous residual connection paradigm by expanding the residual stream width and diversifying connectivity patterns. While yielding substantial performance gains, this diversification fundamentally compromises the identity mapping property intrinsic to the residual connection, which causes severe training instability and restricted scalability, and additionally incurs notable memory access overhead. We propose Manifold-Constrained Hyper-Connections (mHC), a general framework that projects the residual connection space of HC onto a specific manifold to restore the identity mapping property, while incorporating rigorous infrastructure optimization to ensure efficiency.

[LLM-2] Titans: Learning to Memorize at Test Time

Link: <https://arxiv.org/abs/2501.00663>

Over more than a decade there has been an extensive research effort on how to effectively utilize recurrent models and attention. While recurrent models aim to compress the data into a fixed-size memory (called hidden state), attention allows attending to the entire context window, capturing the direct dependencies of all tokens. This more accurate modeling of dependencies, however, comes with a quadratic cost, limiting the model to a fixed-length context. We present a new neural long-term memory module that learns to memorize historical context and helps attention to attend to the

current context while utilizing long past information. We show that this neural memory has the advantage of fast parallelizable training while maintaining a fast inference. From a memory perspective, we argue that attention due to its limited context but accurate dependency modeling performs as a short-term memory, while neural memory due to its ability to memorize the data, acts as a long-term, more persistent memory.

[LLM-3] Differential Transformer

Link: <https://arxiv.org/abs/2410.05258>

Transformer tends to overallocate attention to irrelevant context. In this work, we introduce Diff Transformer, which amplifies attention to the relevant context while canceling noise. Specifically, the differential attention mechanism calculates attention scores as the difference between two separate softmax attention maps. The subtraction cancels noise, promoting the emergence of sparse attention patterns. Experimental results on language modeling show that Diff Transformer outperforms Transformer in various settings of scaling up model size and training tokens. More intriguingly, it offers notable advantages in practical applications, such as long-context modeling, key information retrieval, hallucination mitigation, in-context learning, and reduction of activation outliers. By being less distracted by irrelevant context, Diff Transformer can mitigate hallucination in question answering and text summarization.

[LLM-4] OpenMoE + ReMoE: MoE Routing — Diagnosis and Differentiable Fix

Link 1: <https://arxiv.org/abs/2402.01739>

Link 2: <https://arxiv.org/abs/2412.14711>

OpenMoE: To help the open-source community have a better understanding of Mixture-of-Experts (MoE) based large language models (LLMs), we train and release OpenMoE, a series of fully open-sourced and reproducible decoder-only MoE LLMs, ranging from 650M to 34B parameters and trained on up to over 1T tokens. One important contribution is an in-depth analysis of the routing mechanisms within our OpenMoE models, leading to three significant findings: Context-Independent Specialization, Early Routing Learning, and Drop-towards-the-End. We discovered that routing decisions in MoE models are predominantly based on token IDs, with minimal context relevance. ReMoE: Sparsely activated Mixture-of-Experts (MoE) models scale capacity without increasing computation. The commonly adopted TopK router introduces a discrete and non-differentiable training objective. We propose ReMoE, a fully differentiable MoE architecture that offers a simple yet effective drop-in replacement for the conventional TopK+Softmax routing, utilizing ReLU as the router instead. ReMoE consistently outperforms vanilla TopK-routed MoE across various model sizes, expert counts, and levels of granularity.

[LLM-5] FAC-Synthesis: Less is Enough — Synthesizing Diverse Data in Feature Space of LLMs

Link: <https://arxiv.org/abs/2602.10388>

The diversity of post-training data is critical for effective downstream performance in large language models (LLMs). Many existing approaches to constructing post-training data quantify diversity using text-based metrics that capture linguistic variation, but such metrics provide only weak signals for the task-relevant features that determine downstream performance. In this work, we introduce Feature Activation Coverage (FAC) which measures data diversity in an interpretable feature space. Building upon this metric, we further propose a diversity-driven data synthesis framework, named FAC Synthesis, that first uses a sparse autoencoder to identify missing features from a seed dataset, and then generates synthetic samples that explicitly reflect these features. Experiments show that our approach consistently improves both data diversity and downstream performance on various tasks, including instruction following, toxicity detection, reward modeling, and behavior steering.

MM — Multimodal Learning

[MM-1] The Illusion of Balanced Multimodal Learning: Beyond the Limits of Optimization-Based Methods

Link:

https://drive.google.com/file/d/1vVwJ6ng1k63HhKcmED4gopA_SQRYViD1/view?usp=sharing

Multimodal Sentiment Analysis (MSA) remains constrained by modality imbalance, yet the field continues to rely on optimization-based balancing methods that promise more than they deliver. We provide three contributions: 1) a unified evaluation framework testing gradient and loss-based balancing strategies under controlled settings; 2) a theoretical diagnosis explaining why these methods fail, as they conflate fitting speed with discriminative contribution; and 3) a research agenda toward held-out discriminative modality valuation. Experiments on CMU-MOSI and CMU-MOSEI reveal three shortcomings: no strategy reliably outperforms Late Concatenation; performance is sensitive to hyperparameters; and even ratio calibration fails to yield consistent gains. The core issue is fundamental: loss is not utility, and gradients are not importance. Modality imbalance remains unresolved, motivating utility estimation from held-out performance.

[MM-2] IMDer: Incomplete Multimodality-Diffused Emotion Recognition

Link: <https://github.com/mdswyz/IMDer>

Multimodal emotion recognition suffers from the practical challenge of incomplete modalities, where certain modalities may be missing at test time due to data collection failures or privacy constraints. We propose IMDer, an Incomplete Multimodality-Diffused Emotion Recognition framework that leverages score-based diffusion models to recover missing modality representations conditioned on the available ones. The diffusion-based imputation is more flexible and principled than prior VAE or GAN-based approaches. IMDer achieves state-of-the-art results on IEMOCAP and CMU-MOSI benchmarks under various missing modality scenarios, demonstrating robust emotion recognition even when multiple modalities are unavailable.

RI — Reasoning, Attention and Interpretability

[RI-1] YaRN: Efficient Context Window Extension of Large Language Models

Link: <https://arxiv.org/abs/2309.00071>

Rotary Position Embeddings (RoPE) have been shown to effectively encode positional information in transformer-based language models. However, these models fail to generalize past the sequence length they were trained on. We present YaRN (Yet another RoPE extension method), a compute-efficient method to extend the context window of such models, requiring 10x less tokens and 2.5x less training steps than previous methods. Using YaRN, we show that LLaMA models can effectively utilize and extrapolate to context lengths much longer than their original pre-training would allow, while also surpassing previous state-of-the-art at context window extension. In addition, we demonstrate that YaRN exhibits the capability to extrapolate beyond the limited context of a fine-tuning dataset.

[RI-2] On the Emergence of Thinking in LLMs I: Searching for the Right Intuition

Link: <https://arxiv.org/abs/2502.06773>

Recent AI advancements, such as OpenAI's new models, are transforming LLMs into LRMs (Large Reasoning Models) that perform reasoning during inference, taking extra time and compute for higher-quality outputs. We aim to uncover the algorithmic framework for training LRMs. Methods like self-consistency, PRM, and AlphaZero suggest reasoning as guided search. We ask: what is the simplest, most scalable way to enable search in LLMs? We propose a post-training framework called Reinforcement

Learning via Self-Play (RLSP). RLSP involves three steps: (1) supervised fine-tuning with human or synthetic demonstrations of the reasoning process, (2) using an exploration reward signal to encourage diverse and efficient reasoning behaviors, and (3) RL training with an outcome verifier to ensure correctness while preventing reward hacking. Our key innovation is to decouple exploration and correctness signals during PPO training, carefully balancing them to improve performance and efficiency.

[RI-3] ActivationReasoning: Logical Reasoning in Latent Activation Spaces

Link: <https://arxiv.org/abs/2510.18184>

Large language models (LLMs) excel at generating fluent text, but their internal reasoning remains opaque and difficult to control. Sparse autoencoders (SAEs) make hidden activations more interpretable by exposing latent features that often align with human concepts. Yet, these features are fragile and passive, offering no mechanism for systematic reasoning or model control. To address this, we introduce ActivationReasoning (AR), a framework that embeds explicit logical reasoning into the latent space of LLMs. It proceeds in three stages: (1) Finding latent representations, first latent concept representations are identified (e.g., via SAEs) and organized into a dictionary; (2) Activating propositions, at inference time AR detects activating concepts and maps them to logical propositions; and (3) Logical reasoning, applying logical rules over these propositions to infer higher-order structures, compose new concepts, and steer model behavior. Published as a conference paper at ICLR 2026.

[RI-4] Stop Probing, Start Coding: Why Linear Probes and Sparse Autoencoders Fail at Compositional Generalisation

Link: <https://arxiv.org/abs/2603.28744>

The linear representation hypothesis states that neural network activations encode high-level concepts as linear mixtures. However, under superposition, this encoding is a projection from a higher-dimensional concept space into a lower-dimensional activation space, and a linear decision boundary in the concept space need not remain linear after projection. In this setting, classical sparse coding methods with per-sample iterative inference leverage compressed sensing guarantees to recover latent factors. Sparse autoencoders (SAEs), on the other hand, amortise sparse inference into a fixed encoder, introducing a systematic gap. We show this amortisation gap persists across training set sizes, latent dimensions, and sparsity levels, causing SAEs to fail under out-of-distribution (OOD) compositional shifts. Through controlled experiments that decompose the failure, we identify dictionary learning -- not the inference procedure -- as the binding constraint: SAE-learned dictionaries point in substantially wrong directions. An oracle baseline proves the problem is solvable with a good dictionary at all scales tested. Our results reframe the SAE failure as a dictionary learning challenge,

and point to scalable dictionary learning as the key open problem for sparse inference under superposition.

[RI-5] Physics of Language Models: Part 1, Learning Hierarchical Language Structures

Link: <https://arxiv.org/abs/2305.13673>

Transformer-based language models are effective but complex, and understanding their inner workings and reasoning mechanisms is a significant challenge. Previous research has primarily explored how these models handle simple tasks like name copying or selection, and we extend this by investigating how these models perform recursive language structure reasoning defined by context-free grammars (CFGs). We introduce a family of synthetic CFGs that produce hierarchical rules, capable of generating lengthy sentences (e.g., hundreds of tokens) that are locally ambiguous and require dynamic programming to parse. Despite this complexity, we demonstrate that generative models like GPT can accurately learn and reason over CFG-defined hierarchies and generate sentences based on it. We explore the model's internals, revealing that its hidden states precisely capture the structure of CFGs, and its attention patterns resemble the information passing in a dynamic programming algorithm.

[RI-6] Query-Level Uncertainty in Large Language Models

Link: <https://arxiv.org/abs/2506.09669>

It is important for Large Language Models (LLMs) to be aware of the boundary of their knowledge, distinguishing queries they can confidently answer from those that lie beyond their capabilities. Such awareness enables models to perform adaptive inference, such as invoking retrieval-augmented generation (RAG), engaging in slow and deep thinking, or abstaining from answering when appropriate. These mechanisms are key to developing efficient and trustworthy AI. In this work, we propose a method to detect knowledge boundaries via Query-Level Uncertainty, which estimates if a model is capable of answering a given query before generating any tokens, thus avoiding the generation cost. To this end, we propose a novel, training-free method called Internal Confidence, which leverages self-evaluations across layers and tokens to provide a reliable confidence signal. Empirical studies on both factual question answering and mathematical reasoning tasks demonstrate that our Internal Confidence outperforms several baselines in quality of confidence while being computationally cheaper.

COG — Cognitively Motivated Architectures

[COG-1] Generation is Required for Data-Efficient Perception

Link: <https://arxiv.org/abs/2512.08854>

It has been hypothesized that human-level visual perception requires a generative approach in which internal representations result from inverting a decoder. Yet today's most successful vision models are non-generative, relying on an encoder that maps images to representations without decoder inversion. This raises the question of whether generation is, in fact, necessary for machines to achieve human-level visual perception. To address this, we study whether generative and non-generative methods can achieve compositional generalization, a hallmark of human perception. Under a compositional data generating process, we formalize the inductive biases required to guarantee compositional generalization in decoder-based (generative) and encoder-based (non-generative) methods. We then show theoretically that enforcing these inductive biases on encoders is generally infeasible using regularization or architectural constraints.

[COG-2] Training Language Models via Neural Cellular Automata

Link: <https://arxiv.org/abs/2603.10055>

Pre-training is crucial for large language models (LLMs), as it is when most representations and capabilities are acquired. However, natural language pre-training has problems: high-quality text is finite, it contains human biases, and it entangles knowledge with reasoning. This raises a fundamental question: is natural language the only path to intelligence? We propose using neural cellular automata (NCA) to generate synthetic, non-linguistic data for pre-pre-training LLMs -- training on synthetic-then-natural language. NCA data exhibits rich spatiotemporal structure and statistics resembling natural language while being controllable and cheap to generate at scale. We find that pre-pre-training on only 164M NCA tokens improves downstream language modeling by up to 6% and accelerates convergence by up to 1.6x. Surprisingly, this even outperforms pre-pre-training on 1.6B tokens of natural language from Common Crawl with more compute.

[COG-3] Four Principles for Physically Interpretable World Models

Link: <https://arxiv.org/abs/2503.02143>

As autonomous systems are increasingly deployed in open and uncertain settings, there is a growing need for trustworthy world models that can reliably predict future high-dimensional observations. The learned latent representations in world models lack direct mapping to meaningful physical quantities and dynamics, limiting their utility and interpretability in downstream planning, control, and safety verification. In this paper, we argue for a fundamental shift from physically informed to physically interpretable world models - and crystallize four principles that leverage symbolic knowledge to achieve these ends: (1) functionally organizing the latent space according to the physical intent, (2) learning aligned invariant and equivariant representations of the physical world, (3) integrating multiple forms and strengths of supervision into a unified training process, and (4) partitioning generative outputs to support scalability and verifiability.

SPEC — Special Areas

[SPEC-1]

Link: <https://babylm.github.io/>

BabyLM returns for its **4th year** as both a shared task and a workshop at **EMNLP 2026**. This round keeps the core goal: sample-efficient pretraining under human-scale data budgets, while updating the track structure and datasets.

All data is available [at this huggingface community!](#) Data includes:

- A **detoxified** 100M-word **Strict** dataset and a detoxified 10M-word **Strict-Small** dataset.
- A new **MultiLingual** track based on [BabyBabelLM](#), with evaluation focusing on **English, Dutch, and Chinese**.

Οδηγίες για την εξαμηνιαία εργασία

→ Αναπαραγωγή των αποτελεσμάτων και πιθανές επεκτάσεις ενός ερευνητικού άρθρου στη μηχανική μάθηση μέσω πρακτικού πειραματισμού και ομαδικής συνεργασίας. Οι κύριοι στόχοι είναι: (1) η κριτική επαλήθευση των ισχυρισμών ενός επιλεγμένου άρθρου μέσω πειραμάτων μικρής κλίμακας χρησιμοποιώντας περιορισμένους, ελεύθερα διαθέσιμους υπολογιστικούς πόρους, (2) η διερεύνηση πιθανών επεκτάσεων ή βελτιώσεων πέραν της αρχικής εργασίας, και (3) η συνεργασία σε μια ερευνητική ομάδα μεσαίου μεγέθους.

→ Οδηγίες για τις ενδιάμεσες παρουσιάσεις της εξαμηνιαίας εργασίας. Η παρουσίαση θα γίνει την τελευταία εβδομάδα του εξαμήνου, θα κρατήσει 10 λεπτά και θα συνοδεύεται από διαφάνειες (παρακαλώ να αναρτηθεί ο σύνδεσμος στις διαφάνειες στο google doc δήλωσης εξαμηνιαίας εργασίας) και θα περιέχει:

1. Σύντομη παρουσίαση / σύνοψη του paper που θα υλοποιήσετε (6-8 διαφάνειες)
2. Το πλάνο από τα βήματα που θα ακολουθήσετε (2-4 διαφάνειες)
3. Υπολογιστικοί πόροι και βάσεις δεδομένων που θα χρησιμοποιήσετε (2 διαφάνειες)
4. Με τι θα ασχοληθεί κάθε μέλος της ομάδας (πίνακας 1 διαφάνεια)

→ Θα ακολουθήσουν ερωτήσεις και απαντήσεις (10 λεπτά ανά παρουσίαση). Παρακαλώ να τηρηθούν τα χρονικά όρια.

→ Οδηγίες για τις τελικές παρουσιάσεις της εξαμηνιαίας εργασίας. Η παρουσίαση θα γίνει την τελευταία εβδομάδα της εξεταστικής, θα κρατήσει 15 λεπτά και θα συνοδεύεται από διαφάνειες (παρακαλώ να αναρτηθεί ο σύνδεσμος στις διαφάνειες στο google doc δήλωσης εξαμηνιαίας εργασίας) και θα περιέχει:

1. Πολύ σύντομη παρουσίαση / σύνοψη του paper (1-2 διαφάνειες)
2. Τα αποτελέσματά σας (8-10 διαφάνειες)
3. Αποτίμηση της δουλειάς σας και μελλοντικές επεκτάσεις (1-2 διαφάνειες)
4. Με τι θα ασχοληθεί κάθε μέλος της ομάδας (πίνακας 1 διαφάνεια)
5. Retrospective: Τι μας δυσκόλεψε και θα μπορούσαμε να είχαμε κάνει καλύτερα (1 διαφάνεια)

→ Θα ακολουθήσουν ερωτήσεις και απαντήσεις (5 λεπτά ανά παρουσίαση). Παρακαλώ να τηρηθούν τα χρονικά όρια.

→ Οδηγίες για τελική αναφορά: Η τελική αναφορά θα είναι σύντομη και περιεκτική στην μορφή paper 4-5 δίστηλων σελίδων ή 6-8 μονόστηλων σελίδων. Θα έχει την κλασσική μορφή ερευνητικού paper με έμφαση στην καινοτομία της μεθόδου σας και αναλυτική περιγραφή των αποτελεσμάτων σας.

→ Βαθμολόγηση:

- Τουλάχιστον 2 συναντήσεις με τους επιστημονικούς συνεργάτες 10%
- Ενδιάμεση παρουσίαση 20%
- Τελική παρουσίαση 40%
- Τελική αναφορά 30%

→ Οι εργασίας που ξεχωρίζουν θα πάρουν bonus 10-20%. Κύρια κριτήρια για το bonus είναι η καινοτομία, ερευνητικά αποτελέσματα πέραν αυτών που αναφέρονται στο paper που υλοποιήσατε και η καλή συνεργασία ανάμεσα στα μέλη της ομάδας.

→ Σημαντικά θέματα:

- Η συνεργασία ανάμεσα στα μέλη της ομάδας σας δεν θα είναι πάντα εύκολη λόγω του μεγέθους της ομάδας σας, διαφορετικών ωραρίων και υποχρεώσεων. Πέραν των συναντήσεων, είναι σημαντικό να έχετε τακτική επικοινωνία, π.χ., μέσω slack ή αντίστοιχου εργαλείου, όπου θα περιγράφετε την καθημερινή σας πρόοδο σε όλα τα μέλη της ομάδας.
- Πρέπει να είσαστε πολύ προσεκτικοί ως προς το μέγεθος των μοντέλων και των δεδομένων που θα χρησιμοποιήσετε γιατί δεν θα σας δοθούν υπολογιστικοί πόροι. Οπότε από την αρχή πρέπει να είστε σίγουροι ότι μπορείτε να τρέξετε τα πειράματά σας με δωρεάν πόρους όπως το google colab (<https://colab.research.google.com/>)

Instructions in English

→ Goal: Reproduction of results and possible extensions of a research paper in machine learning through hands-on experimentation and team collaboration. The main objectives are: (1) critical verification of the claims of a selected paper through small-scale experiments using limited, freely available computational resources, (2) exploration of potential extensions or improvements beyond the original work, and (3) collaboration in a mid-sized research team.

→ Guidelines for the interim presentations of the semester project: The presentation will take place during the last week of the semester, will last 10 minutes, and must be accompanied by slides (please upload the link to your slides in the Google Doc where you declared your semester project). It should include:

1. A brief presentation/summary of the paper you will implement (6-8 slides)
2. A plan outlining the steps you will follow (2-4 slides)
3. Computational resources and datasets you will use (2 slides)
4. A breakdown of what each team member will be working on (table, 1 slide)

→ A Q&A session will follow (10 minutes per presentation). Please respect the time limits.

→ Guidelines for the final presentations of the semester project: The final presentation will take place during the last week of the exam period, will last 15 minutes, and must be accompanied by slides (please upload the link to your slides in the Google Doc where you declared your semester project). It should include:

1. A very brief presentation/summary of the paper (1-2 slides)
2. Your method and results (8-10 slides)
3. Evaluation of your work and potential future extensions (1-2 slides)
4. A breakdown of what each team member worked on (table, 1 slide)
5. Retrospective: What challenged us and what we could have done better (1 slide)

→ A Q&A session will follow (5 minutes per presentation). Please respect the time limits.

→ Final Report Guidelines: The final report should be concise and in the format of a research paper: 4-5 pages (two-column layout) or 6-8 pages (single-column layout). It should follow the typical structure of a research paper, with an emphasis on the innovation of your method and a detailed description of your results.

→ Grading:

- At least 2 meetings with academic supervisors: 10%
- Interim presentation: 20%
- Final presentation: 40%
- Final report: 30%

→ Outstanding projects will receive a bonus of 10-20%. The main criteria for the bonus are: innovation, research results that go beyond what is presented in the implemented paper, and good collaboration among team members.

→ Important Notes:

- Collaboration among team members may not always be easy due to the size of the team, varying schedules, and other obligations. Beyond scheduled meetings, it's important to maintain regular communication - e.g., via Slack or a similar tool - where everyone can share their daily progress with the whole team.
- You must be very careful about the size of the models and datasets you use, as no computational resources will be provided. Therefore, from the start, ensure you can run your experiments using free resources such as Google Colab (<https://colab.research.google.com/>).